Carisurg

Assignment 21

Final Handover Documentation

Due Date: July 28th, 2026.

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Project Summary

This repo trains and evaluates a machine-learning model that flags whether a patient may be ESI-1 (the most critical triage level) from ten triage-time features (age, vitals, chief-complaint flags, arrival mode) — a screening aid intended to run alongside a nurse's own triage judgement, not a replacement for it.

It was built on a public Yale EMMLC dataset (55,121 encounters) as a proof of feasibility while the Phase 1 Caribbean data-capture tool is developed in parallel; it has **not** been validated on Mercer General data and is not cleared for clinical use.

The Final Model Decision

The model used is **Logistic Regression**, `class_weight="balanced"`, pinned in `config.yaml`.

Deployed as an ESI-1 screening flag alongside normal triage — not as an autonomous ESI classifier.

A nurse still assigns ESI 2–5 exactly as she does today; this model's only role is to flag "is this patient one of the critical ones you might be missing?" That framing matters: at 0.254 overall accuracy, this model cannot defensibly assign every patient's full ESI level on its own (Dr. De Freitas raised this directly — see the Week 8 addendum in the decision journal linked below). It was never evaluated for that job, and is not deployed for it.

Why?

Across six models tried in Weeks 6–7 (Dummy, Decision Tree, Logistic Regression, Random Forest, tuned Random Forest, Gradient Boosting), Logistic Regression has the lowest accuracy but by far the highest recall on ESI-1 — the most critical patient class (0.750, versus 0.000 for Random Forest and 0.375 for Gradient Boosting).

For a triage system, missing critical patients is a worse failure than a lower overall accuracy score.

- Full **comparison**: `modules/week-08/docs/week08_selection.md`
- Full **reasoning**: `modules/week-08/docs/decisions/week08_choice.md`

How to Run?

```bash
git clone https://github.com/j0hnc0d3s/carisurg.git
cd carisurg/modules/week-08
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# placed triage_cleaned_v1.csv in dataset/ (see "Where the Data Lives" below)

python scripts/train.py --config config.yaml
```

Expect console output ending in a metrics block (accuracy, ESI-1 recall, etc.) and three new files under `records/`: `final_model.joblib`, `final_scaler.joblib`, `final_metrics.json`.

To confirm the pipeline itself is healthy before trusting any of that:

```bash
python -m pytest -v
```

Both tests should pass in a few seconds. They check the data schema and run a training smoke test on synthetic data — they do not check model quality.

Where the Data Lives?

1. **Source**: Yale Emergency Medicine Machine Learning Consortium (Yale EMMLC), a US academic hospital's de-identified ED triage export.
2. **File expected at**: `dataset/triage_cleaned_v1.csv` — git-ignored, not committed to this repo. See `modules/week-08/README.md` for provenance and how to obtain it.
3. **Access**: limited to CariSurg MedTech Pathways programme participants and tutors. Do not redistribute the CSV outside the programme, upload it to any third-party service, or commit it (or any copy/derivative of it, including model artefacts trained on it) to a public repository.
4. **Governance status**: external, de-identified, public research dataset — not Mercer General patient data, and no IRB/ethics approval has been sought or is required for it. De-identified does not mean ungoverned: the access restriction above still applies. Any future work using real Mercer General patient data will require separate ethics review before this pipeline (or any successor) touches it.

Known Limitations

1. **Overall accuracy (0.254) is not acceptable for autonomous ESI assignment — this is why the model is scoped to an ESI-1 screening flag, not a full classifier (see decision above).** Deploying it to assign every patient's ESI level unsupervised would mean it's wrong on roughly 3 of 4 patients across ESI 2–5, which risks both direct triage errors on the common classes and clinicians losing trust in the tool. If a future phase wants a full ESI classifier, that is a different, harder problem than this one, and should not inherit this model's scope by default.
2. ESI-1 recall is measured on only 16 test patients (77 of 55,121 encounters overall). Every recall number in `modules/week-08/docs/week08-selection.md` should be read as a consistent pattern across models, not a precise estimate — a single patient's outcome shifts the ESI-1 recall figure by more than 6 percentage points.
3. This is not Caribbean data. No result in this repo says anything about performance at Mercer General until validated against local ED data collected under Phase 1. Single-site data of any kind carries a real risk of distribution shift when applied elsewhere.
4. The pinned model uses only 10 features, and demographics are excluded by design, not oversight.
5. No race, ethnicity, or other demographic column is in the feature set — this was a deliberate fairness choice from the Week 5 memo, not a gap to fill in later. It does mean the model draws on far less than a clinician's full judgement, which caps how good it can realistically get.

Who to Ask?

Question About	Contact
Model Choice	Josiah-John Green
Dataset Access	Martina Griffith (Clinical IT Lead)
Clinical Validity of ESI Predictions/Triage Workflows	Dr Marcus Reyes (Consultant EP) or Dr De Freitas (Clinical Sponsor)
Programme Process/Submission Logistics	Carisurg Tutor Team, in '#ask-a-tutor'

Appendices

Supplementary Links

1. The GitHub link can be found <u>here</u>
2. The decision journal can be found in the docs for this week.

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as (SBP + 2×DBP) / 3
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	**Meaning**
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. "admit") and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	**Meaning**
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol